



CAPR: Confidence-Aware Prompt Refinement in Large Language Models

Jen-Tzung Chien^{1,2}, Po-Chun Huang²

¹Inst. of Electrical and Computer Engineering, National Yang Ming Chiao Tung University, Taiwan

²Institute of Computer and Comms. Security, National Yang Ming Chiao Tung University, Taiwan

jtchien@nycu.edu.tw, qmao.cs11@nycu.edu.tw

Abstract

Mitigating hallucination in large language models (LLMs) is crucial to ensure trustworthy generation. A meaningful solution to tackle this issue involves eliciting a reliable confidence score in the generation. However, the previous methods only focused on using white-box models for short-form generation. The solutions to long-form sentences with black-box models remain very limited. This study proposes a reliable black-box confidence elicitation for long-form generation by leveraging the external knowledge, inspired by retrieval augmented generation. In particular, this paper proposes the confidence-aware prompt refinement (CAPR) to modify user prompts to ensure trustworthy responses. The prompt refiner is trained by maximizing the rewards based on the confidence and the accuracy of generated sentences. Experiments show that CAPR effectively elicits reliable confidence scores. The refined prompts enable LLMs to produce reliable responses with calibrated confidence.

Index Terms: large language model, prompt refinement, hallucination mitigation, question answering

1. Introduction

The ability of the current open-domain dialogue system, empowered by large language models (LLMs), has changed the user behaviors in utilizing natural language applications. The open-domain dialogue system, which was initially designed for chit-chat purpose, has been transformed into a general system that can handle various users' queries. For example, instead of asking search engines about factoid questions, people currently use an open-domain dialogue system like ChatGPT [1] and expect to find the helpful and right answer. Unfortunately, such kind of system does not always provide reliable answers to individual queries due to the issue of hallucination [2, 3]. Hallucinated answers generally refers to those misleading responses that are presented in a confident and seemingly plausible manner. This issue does exist because of twofold reasons. The first one is the inherent limitations in LLMs, resulting in insufficient knowledge during training, while the second one is the quality of the input prompts at test time.

To address the limitation in LLM capability, some recent studies have presented the mechanisms to elicit a confidence score indicating the level of model certainty for a response generation [4]. In a white-box setting, such a score was measured by accessing an internal information of token likelihood from a visible and tunable model. However, this approach is infeasible to black-box models, like ChatGPT, due to the inaccessibility of internal model information. A straightforward method to elicit confidence scores from black-box LLMs involves a verbal elicitation by using a prompt [5]. Unfortunately, some prior findings have demonstrated that black-box LLMs tend to be overconfi-

dent in their elicited scores. To mitigate generation of overconfident responses, verbal confidence elicitation was performed by aggregating several confidence scores [6–8]. This approach, however, required multiple prompting steps for a single query in an aggregation procedure, leading to a time-consuming prompt strategy. Another issue with the previous works is that the majority of the solutions were tailored only for short-form generation, such as question-answering (QA) tasks [9–12], without considering long-form responses [13, 14].

In addition, the limitation of prompt quality [15, 16] was tackled by designing the sophisticated prompt strategies to enhance the trustworthiness in LLM responses. A notable solution was based on the in-context learning (ICL) [17]. However, this advanced prompt strategy is primarily feasible to short-form generation tasks. Long-form generation, such as a long-form answering system, poses a unique challenge to deal with the scenario of searching the extended and detailed answers from a short input question with very limited information. The utilization of ICL to cope with this scenario did not yield significant improvement because the in-context demonstrations were usually too lengthy, potentially confusing the generation of answers. Additionally, the extensive input tokens for such demonstrations consume a large token capacity within the black-box model, which results in an expensive prompting process.

This paper addresses both limitations and proposes two solutions. Firstly, a knowledge-based confidence elicitation is introduced to ensure a regularized confidence score from black-box LLMs for a reliable long-form generation. In lieu of merely averaging scores from multiple black-box models, this approach leverages external knowledge to calibrate the confidence score, drawing the inspiration from retrieval-augmented generation (RAG). Secondly, the elicited confidence scores are utilized to train a prompt refiner, based on the so-called confidence-aware prompt refinement (CAPR) model, to address the challenge of the limited prompting options in long-form generation tasks. The proposed method is evaluated on the challenging long-form QA task using different black-box LLMs, demonstrating the advantages of the suggested approaches.

2. Confidence Elicitation & Prompt Refinement

2.1. Confidence elicitation in large language models

Confidence elicitation aims to provide a regularized confidence score to reflect truth and correctness of an LLM response, thereby helping users gauge the trustworthiness. The related methods are categorized into white-box and black-box approaches. White-box methods commonly leverage the model's internal mechanisms using the controllable LLM, such as prob-

ability consistency and weight stability during inference for token generation [12, 18]. Conversely, black-box methods, which interact solely with the inputs and outputs of the LLM, and elicit the confidence scores verbally by incorporating an input prompt for confidence score generation. However, prior works [6–8] have demonstrated that black-box LLMs often exhibit too high score, typically ranging between 0.8 and 1 (the highest score), and accordingly generate overconfident responses. This discrepancy reflects the unreliability in black-box confidence scores, particularly in the highly-secured applications such as healthcare and finance, where accurate assessments are vital. To address this issue, the methods to LLM responses such as using multiple sampling for consistency estimation or employing external white-box model to generate reliability estimates have been proposed [6–8]. These approaches, however, lead to costly prompting as the confidence score elicitation needs the repeated prompting. Different from the previous works, this study introduces the knowledge-based confidence elicitation by utilizing external knowledge based on information retrieval to reliably elicit confidence scores at a reduced cost.

2.2. Prompt refinement to mitigate hallucination

The hallucination issue in LLMs can be mitigated through the hand-crafted prompt engineering techniques, which are required for trustworthy generation. For example, the so-called rephrase and rewrite (RaR) in [19] employed a hard prompt template for prompt refinement where an input question is augmented with a fixed hard prompt in a form of

{Question} Rephrase and expand the question, and respond.

This method stimulates LLMs to fulfill a reasoning process via rephrasing and rewriting, and empirically outperformed the traditional prompt strategies in various tasks. Additionally, the prompt strategy based on a Self-Polish method [20] took the advantage of LLM for language understanding where an iterative refinement procedure was performed through rewriting the questions to enhance the prompting performance. On the other hand, an alternative prompt strategy called the TextGrad [21] method introduced the textual-gradient descent, which was conceptually inspired by the backpropagation scheme to optimize the prompts given by LLMs. In general, the aforementioned methods were proposed to improve the quality of responses as well as increase the reliability of LLM outputs.

3. Confidence-Aware Prompt Refinement

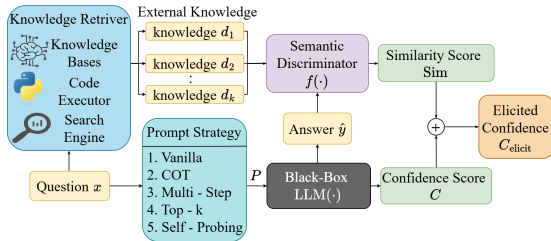


Figure 1: Overview of knowledge-based confidence elicitation.

3.1. Knowledge-based confidence elicitation

This study presents a knowledge-based confidence elicitation (KCE) framework to improve the alignment between confi-

dence score and generation accuracy by using black-box LLMs like GPT-3.5 and GPT-4. This KCE method is implemented by assessing the semantic similarity between generated responses and factual knowledge. As shown in Figure 1, given an input question x , this framework starts by generating a set of initial responses $\hat{y}$ along with the corresponding verbalized confidence scores C based on different prompt strategies P . The system outputs $\hat{y}$ are evaluated by LLM to determine scores $C = \text{LLM}(x, P)$. At the same time, the external knowledge from k documents is retrieved via a retriever $\{d_j\}_{j=1}^k = \text{Ret}(x)$ by using question x . Then, the knowledge-based confidence elicitation $\text{KCE}(\cdot)$ is determined by calibrating the confidence score based on information sources from multiple prompts and multiple documents, referring to retrieval augmented generation, via

$$C_{\text{elicit}} = \lambda \cdot C + (1 - \lambda) \cdot \text{Sim} \quad (1)$$

where λ is used to balance the influences from the verbalized confidence C and the similarity score, computed by $\text{Sim} = f(\hat{y}, \{d_j\}_{j=1}^k) = \max_{1 \leq j \leq k} \left(\frac{\hat{y} \cdot d_j}{|\hat{y}| |d_j|} \right)$, which is measured by a semantic discriminator $f(\cdot)$ between knowledge documents $\{d_j\}_{j=1}^k$ and current answer $\hat{y}$. The confidence score C is therefore penalized with low similarity. This calibration would like to regularize LLM outputs for trustworthy generation.

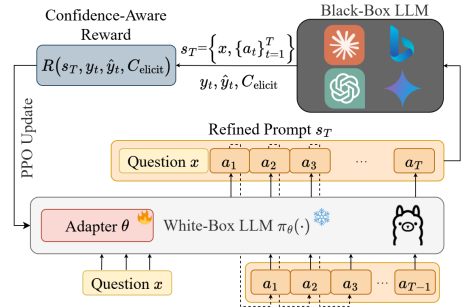


Figure 2: Overview of confidence-aware prompt refinement.

3.2. Confidence-aware adapter for prompt refinement

In addition to confidence calibration [22, 23], this paper further mitigates hallucination in LLM through a reinforced prompt refinement driven by a specialized reward function [24, 25]. A new method called confidence-aware prompt refinement (CAPR) is proposed and implemented as depicted in Figure 2. This method leverages the output of KCE to calculate a reward function for training a reinforcement learning (RL) agent to refine prompt and update answer $\hat{y}$. By fulfilling RL with human feedback (RLHF), an optimization process is formulated as a Markov decision process. Given a query or question x , a_t is defined as a token generated by RL agent at time step t . Notably, the state in RL is defined as an augmented prompt input $s_t = \{x, a_0, \dots, a_t\}$. This state at the end of the trajectory, s_T , is known as the final refined prompt. Accordingly, the RL agent is built by using a pre-trained white-box LLM, where the only trainable parameters are the adapter layers with parameter θ . The optimization process in CAPR is specially designed for long-form generation for question answering.

3.2.1. Confidence-aware reward function

Reward function r_t plays a crucial role when fine-tuning an LLM with RL using adapter θ . The reward is basically designed

to assess the quality and reliability of the generated answer $\hat{y}$ from a query x . It can be calculated by using the expected calibration error (ECE) and the generation accuracy (ACC) as

$$r_t \triangleq R(s_T, \hat{y}_t, y_t, C_{\text{elicit}}) = -\text{ECE} + \text{ACC} \quad (2)$$

where y is the ground truth response, $\hat{y}$ is the response with the refined prompt s_T , and C_{elicit} is the elicited confidence from KCE at time step T . Optimizing this function involves minimizing the ECE and maximizing the accuracy, likely leading to reliable and accurate outputs [26, 27]. However, the definition of accuracy may vary over different downstream tasks.

Following standard RLHF with human feedback y , such a reward $R(\cdot)$ is only calculated at the end of trajectory $\tau = \{\tau_t\} = \{s_T, \hat{y}_t, y_t, r_t, C_{\text{elicit}}\}$ at time step T , which is quite similar to the scenario in the contextual bandit algorithm. In particular, ECE is defined by measuring the accumulated distance between the accuracy and the confidence over the model predictions across M bins with each bin having the length $|B_m|$

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{Accuracy}(B_m) - \text{Confidence}(B_m)| \quad (3)$$

where n denotes the number of QA training samples within a bin B_m . Predictions are sorted into these bins based on their confidence scores, and the ECE is computed as the weighted average of the absolute differences between the accuracy and the confidence for different samples in individual bins.

The confidence-aware reward is subsequently computed and utilized for training a prompt refiner [28] via adapter layer θ within the white-box LLM by employing the proximal policy optimization (PPO) algorithm [29]. This trick can avoid fine-tuning LLMs which is time-consuming. The refined prompt s_T is formed by transforming original question x with the sequence of tokens generated by the policy network π_θ , where each generated token can be formulated as $a_t \sim \pi_\theta(a_t|x, a_0, \dots, a_{t-1})$. Since $\hat{y}$ and C are generated by black-box LLMs, the proposed learning algorithm follows the implementation based on the hybrid black-box and white-box LLMs where black-box LLM is used to provide the calibrated confidence C_{elicit} and white-box LLM is employed to train a prompt refiner θ via continuous learning with RL, which can be fitted to long-form answer $\hat{y}$.

3.2.2. Adapter learning for prompt refinement

PPO is implemented as an RL solution to optimize the adapter or policy π_θ by maximizing the expected reward [30]. The policy gradient method is applied for optimization. In particular, the learning objective is to maximize the expected reward

$$\mathbb{E}[R(s_T, \hat{y}_t, y_t, C_{\text{elicit}})] = \int R(s_T, \hat{y}_t, y_t, C_{\text{elicit}}) \pi_\theta(a_t|s_t) da_t. \quad (4)$$

The optimization problem is formulated by maximizing

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} [R(\tau)] \quad (5)$$

where the refined prompt is sampled from CAPR trajectory τ leveraged by KCE. Notably, the policy gradient is derived by

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a_t|s_t) R(s_T, \hat{y}_t, y_t, C_{\text{elicit}})]. \quad (6)$$

Following the PPO process, which incorporates a clipped surrogate objective to constrain the magnitude of policy updates

with an advantage function [29], ensuring that these updates stay closely aligned with the previous policy, the optimal policy parameter is estimated by

$$\theta^* = \arg \max_{\theta} \sum_{\pi_\theta(x)} \mathbb{E}_{\tau \sim \pi_\theta} [R(s_T, \hat{y}_t, y_t, C_{\text{elicit}})]. \quad (7)$$

By optimizing the prompt refiner θ with a carefully designed confidence-aware reward function, the proposed CAPR is trained via a gradient ascent algorithm for a fine-tuned LLM

$$\theta_{\text{new}} \leftarrow \theta_{\text{old}} + \alpha \nabla_\theta J(\theta)|_{\theta=\theta_{\text{old}}} \quad (8)$$

using a learning rate α . Finally, the optimal prompt is refined and sampled by the optimal policy $a_t^* \sim \pi_{\theta^*}(a_t|s_t)$. CAPR is implemented for trustworthy generation as Algorithm 1.

Algorithm 1: Training procedure for prompt refiner

Require: initial policy θ_0 , retriever function $\text{Ret}(\cdot)$, reward function $R(\cdot)$, knowledge confidence elicitation KCE($\cdot$) and prompt refiner π_θ

initialize $\theta^* = \theta_0$

repeat

 sample question and ground truth pair (x, y)

 retrieve k documents $\{d_j\}_{j=1}^k = \text{Ret}(x)$

for each step t **do**

 generate prompt token by $a_t \sim \pi_\theta(x)$

end

for each step t **do**

 generate answer $\hat{y}_t$ and elicit confidence

$C_{\text{elicit}} = \text{KCE}(\{d_j\}_{j=1}^k, x, \{a_t\}_{t=0}^T)$

 compute reward function

$r_t = R(s_T, \hat{y}_t, y_t, C_{\text{elicit}})$

 collect trajectory with transition

$\tau_t = \{s_t, \{a_t\}_{t=0}^T, r_t, s_{t+1}, C_{\text{elicit}}\}$

end

 update prompt refiner $\pi_\theta(\cdot)$ using PPO with τ

until convergence or max iteration

return θ^*

4. Experiments

4.1. Experimental setup

4.1.1. Backbone and prompt refiner models

This study employed GPT-3.5 and GPT-4 as black-box LLMs in the experiments due to their superior performance on the previous reasoning benchmarks such as MMLU, GLUE, SQuAD, and RACE. Furthermore, these models are among the most widely utilized black-box LLMs and have been applied in numerous applications. For the prompt refinement model, this study employed the adapters configured in white-box model based on LLaMA2 [31], which underwent fine-tuning by using the PPO algorithm to enhance the user's initial prompts, thereby resulting in improving the responses, which were generated by the black-box LLMs. The efficient fine-tuning of the prompt refiner was proposed by utilizing a sophisticated confidence-aware reward function and adapter layers. LLaMA2-adapter architecture was the same as that in [31]. $\lambda = 0.8$ was used.

4.1.2. Experimental dataset

The ASQA (answer summaries for questions which are ambiguous) dataset [32] was selected as the benchmark for long-form

QA task. This dataset is a well-established resource in the field of closed-book, open-domain QA tasks, providing a comprehensive collection of real-world questions and their corresponding answers in a detailed and paragraph-length style across a vast range of topics. The prompt strategies using vanilla, chain-of-thought (COT) [33], multi-step [34], top- k sampling and self-probing were considered. Given the ambiguous questions, the KCE was able to elicit accurate confidence scores from black-box LLMs. Concurrently, the CAPR was skilled at revising ambiguous questions into more precise ones, thereby guiding the black-box LLMs to produce better outputs.

4.1.3. Evaluation metrics

Several evaluation metrics were employed to assess the quality of the proposed method. ROUGE-L [35] measures the accuracy by comparing the longest common subsequence between generated and ground truth texts. Consistent with the previous study setting [12], this work defines ROUGE-L score as the accuracy metric for long-form QA, using a threshold of 0.3. ECE evaluates how well the model’s predicted probabilities align with the actual outcomes, while the area under the receiver operating characteristic curve (AUROC) gauges the model’s ability to predict confidence accurately in classifying responses. Generally speaking, these metrics provide a comprehensive evaluation of the model’s accuracy, trustworthiness, and prediction capability.

Table 1: Performance comparison of different long-form generation methods on different black-box LLMs.

	ROUGE-L ($\uparrow$)	ECE ($\downarrow$)	AUROC ($\uparrow$)
GPT-3.5			
Original LLM	0.1821	0.7598	0.5512
Self-Polish [20]	0.2163	0.7316	0.5125
RaR [19]	0.1423	0.7746	0.5631
TextGrad [21]	0.2137	0.7222	0.6625
CAPR w/ KCE	0.2377	0.6359	0.6688
GPT-4			
Original LLM	0.1945	0.7562	0.6002
Self-Polish [20]	0.2121	0.7452	0.5833
RaR [19]	0.2407	0.7010	0.5203
TextGrad [21]	0.2155	0.6826	0.5942
CAPR w/ KCE	0.2648	0.6466	0.8065

Table 2: Ablation studies on the benefit of KCE in CAPR with different prompt strategies and black-box LLMs.

	Vanilla	COT	Multi-step
ECE ($\downarrow$)			
GPT-3.5	0.7082	0.7778	0.7548
GPT-3.5 w/ KCE	0.5906	0.5260	0.5975
GPT-4	0.7327	0.8950	0.7961
GPT-4 w/ KCE	0.6086	0.6932	0.6048
AUROC ($\uparrow$)			
GPT-3.5	0.5255	0.5440	0.5039
GPT-3.5 w/ KCE	0.6071	0.7268	0.6940
GPT-4	0.5972	0.5769	0.4873
GPT-4 w/ KCE	0.6437	0.8622	0.7529

4.2. Experimental results

The results shown in Table 1 demonstrate the performance of different prompt refinement methods, including Self-Polish [20]

Table 3: Ablation studies on reward functions

	ROUGE-L ($\uparrow$)	ECE ($\downarrow$)	AUROC ($\uparrow$)
CAPR w/o KCE	0.1808	0.6334	0.51113
CAPR w/ ACC	0.2151	0.7679	0.5340
CAPR w/ KCE	0.2226	0.6042	0.5777

and TextGrad [21], in generating reliable prompts for long-form QA tasks. Original LLM is the case of standard prompt. Self-Polish leveraged the LLM to iteratively improve the question, while Textgrad employed textual gradient descent with GPT-4 to align responses close to ground truth answers. Another method, rephrase and respond (RaR), used a structured prompt to simultaneously rephrase and answer the question. The proposed approach notably enhances the ROUGE-L, ECE and AUROC scores, boosting the reliability and accuracy of confidence assessments in LLM outputs. This set of experiments shows that CAPR with KCE performs well in different black-box LLMs, emphasizing its generalization performance. This improvement is crucial for applications that depend on precise and dependable model responses, enhancing the user trust and the utility of the generated content. The results using GPT-4 are consistently better than those using GPT-3.5.

4.3. Ablation studies

To demonstrate the effectiveness of the proposed methods, ablation studies are conducted to evaluate the knowledge-base confidence elicitation (KCE) and the reward function within CAPR. Table 2 shows the evaluation on the effect of KCE in different traditional prompt strategies, showing that incorporating KCE consistently leads to more reliable confidence scores and better generation results in terms of ECE and AUROC. In Table 3 with the results on ROUGE-L, ECE and AUROC, CAPR without KCE represents the case of using CAPR with only verbalized confidence score C while CAPR with ACC implies CAPR using accuracy as the sole reward function. CAPR with KCE is the proposed method using C_{elicit} . This comparative study on reward function highlights the synergistic relationship between KCE and CAPR, demonstrating that each component enhances overall performance. The analysis clearly shows how KCE complements CAPR and significantly improves prompt generation for long-form answering in different metrics.

5. Conclusions

This paper introduced two complimentary approaches to dehalucination in black-box LLMs. These two methods were integrated to strengthen the solution by using the external knowledge to calibrate LLM outputs, and cross-referencing them with reliable prompt to improve reliability of response. The hybrid solution ensures that it is likely to generate factual responses given by the grounded prompt. The prompt refinement was iteratively performed by fine-tuning LLMs according to a prompt generator based on reinforcement learning, using the feedback from black-box LLMs and helping the generation of reliable and contextually relevant answers over time, which were fitted to the applications of long-form generation. By integrating the confidence calibration and the prompt refinement, the proposed method enhanced the trustworthiness of black-box models such as GPT-3.5 and GPT-4 over ASQA task, minimizing the hallucinations while simultaneously improving both accuracy and reliability of their responses.

6. References

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